

Mid-semester examination, M.Sc. Semester 2
(Finite automata and compiler design)
Full Marks - 20
Time - 1 hour

Answer any four questions.

[4 x 5 marks = 20 marks]

1. Write the formal definition of regular grammar. Give the regular grammar for the language $L = \{0^n \mid n \geq 1\}$ over $T = \{0\}$.
2. Design a DFA that accepts an odd number of 1s.
3. What is an ambiguous grammar? How can you eliminate ambiguity? Explain with an example. Justify also.
4. Name and explain the different error recovery strategies used in parsing.
5. Write quadruples, triples, and indirect triples for the expression:

$$(a*b) + (c+d) - (a+b+c+d)$$